MGIS 355 · BUSINESS INTELLIGENCE · FALL 2026

Project 1 — Data model and governed analysis

Assigned Thursday, September 17 · checkpoint Thursday, October 1 · due Sunday, October 18

A ranking of fourteen stores against their quarterly targets, built on a model you designed, from metrics you defined, with a record of how you checked it. An agent can produce a store ranking in ninety seconds. This project is graded on everything that makes the ranking worth handing to someone.

WEIGHT	20% of the course grade
CHECKPOINT	Thursday, October 1, in class — required, graded on completion
DUE	Sunday, October 18, 11:59 pm, on MyCourses
WORK	Individual
AI TIER	AI-Open. Use any agent you like. Disclose what it did and verify what it produced.
DATA	`ridgeline.duckdb`, the Ridgeline data dictionary, and `queries_to_audit.sql` — all on MyCourses

1 · The situation

You are the analyst for Ridgeline Outfitters, a fourteen-store outdoor retailer. This memo is sitting in your inbox.

FROM THE VP OF STORES · SUBJECT: Q3 STORE RANKING FOR THE DISTRICT REVIEW

I have the district review on the 22nd and I need to know which stores are actually behind. Marcus in Burlington tells me his numbers look bad because everything he sells now gets picked up instead of rung up, and the credit goes somewhere else. Two of my district managers are quoting me different revenue figures for the same store.

Give me a ranking of the fourteen stores against their Q3 targets, tell me which ones are genuinely behind, and tell me what you did about Marcus's complaint. I want one page I can hand out, and I want to be able to defend every number in it if a store manager pushes back.

Marcus is not wrong. Buy-online-pick-up-in-store went from a twentieth of Ridgeline's orders to nearly a fifth in a year, and the extract can attach a store to an order through three different columns.

6.2%
of 2025 orders were picked up in a store

19.2%
of 2026 orders were picked up in a store

3
columns that can attach a store to an order

THE SHORT VERSION

Whichever rule you choose for crediting a store now moves three times as much money as it did last year. That is a definition rather than a data-cleaning problem, and two reasonable analysts will choose differently. Producing the ranking is the cheap part — you are graded on the grain, the definitions, the verification, and whether you can defend the number when a store manager pushes back.

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2 · What you are building

A small dimensional model of Ridgeline's selling process, a set of formally defined metrics on top of it, and a one-page recommendation supported by a verification record.

You are **not** building a dashboard, a Streamlit app, or a predictive model. That is Project 2. Extra scope will not earn extra credit.

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3 · The five decisions you have to make and defend

These are the project. Everything else is mechanics.

1. **Store credit.** Which store gets credit for an order? `selling_store_id`, `pickup_store_id` and `returns.return_store_id` each answer differently, and each is blank for some orders. Web orders that ship have no store at all. Pick a rule, state it, and say what it does to Marcus.
2. **What "revenue" means.** Gross or net of returns. Discounts exist at two grains and are never reconciled with each other. Shipping charged to the customer may or may not be revenue. Tax. Cancelled orders. Each choice is defensible; an undeclared choice is not.
3. **Grain.** One sentence per fact table, written *before* you write the query that builds it. Returns are recorded per line; order status is recorded per order. They cannot be made to agree, so you have to decide which one your measure of "returned" follows.
4. **Calendar.** Timestamps are UTC and the stores are Eastern. Which date a sale falls on depends on which column you build on and which timezone you resolve it in. Some sales change quarter.
5. **Comparability.** The extract holds a full Q3 2025 and 2026 orders only through August 31. Targets are one number per store per fiscal quarter. Comparing a partial quarter against a whole-quarter target requires a rule you invent and state.

"Can you say what you chose, why, and what the alternative would have done to the answer?"

That is the question asked of each of the five, and it is not the same question as "did you choose correctly." There is no answer key for a definition.

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4 · Deliverables

Submit one folder, zipped, named `mgis355_p1_lastname_firstname`.

1 · Decision memo — `memo.pdf`, two pages maximum

Page one is the deliverable the VP asked for: the ranking, which stores are genuinely behind, your answer on store credit, and what you recommend. Written for a VP, not for me. No code, no table of contents, no apologies about the data.

Page two is the framing, in the five-field form from Session 2 — what decision, who makes it, what would change it, what metric, what threshold — plus your five decisions above, one or two sentences each, with the alternative you rejected.

2 · The model — `model/`

- **A bus matrix.** One row per fact table, one column per dimension, a mark where they connect. A table in a markdown file is fine.
- **A grain statement for each fact table.** One sentence, in the form "one row per ___."
- **A build script** — `build_star.py` or `build_star.ipynb` — that reads from `ridgeline.duckdb` and writes your star to a separate database file. It must run start to finish on a clean machine and produce the same result twice. I will run it.

Minimum scope: at least two fact tables at different grains, at least three dimensions, one dimension used in more than one role, and a date dimension carrying a local business date.

3 · Metric definitions — `metrics.md`

Every metric you quote anywhere in the submission gets an entry with these fields:

FIELD	WHAT GOES IN IT
Name	The name a store manager would use
Formula	In terms of your star's columns, not prose
Grain	The lowest level at which it is valid
Filters	Including the ones you chose not to apply
Store credit rule	Which of the three attributions it uses
Calendar	Which date column, and which timezone
Owner and status	Who would own this definition at Ridgeline, and whether you consider it settled or provisional

At least four metrics, including one rate — a ratio of two measures — and one comparison against target. If two metrics differ only by a filter, they are two metrics and need two entries.

4 · Analysis notebook — `analysis.ipynb`

Reads from your star with SQLAlchemy and `pd.read_sql`. Parameterized queries — no values formatted into query strings. Aggregation belongs in SQL unless you can say why it belongs in pandas. One step per cell, and a sentence above each cell saying what the step is for. Charts only where a chart earns its place; this is not a visualization project.

5 · Verification log — `verification.md`

The section this project is actually about. Three parts:

- **Three traced numbers.** Pick three figures that appear in your memo and trace each one back to the source extract by a route that does not go through your star. Show the query or the cells, the number you got, and whether it matched. If it did not match, say so and explain why — a documented mismatch scores better than a silent one.
- **One audited query.** Take a query you did not write — one an agent generated for you, or one from `queries_to_audit.sql` — and audit it. State what it claims to return, what it actually returns, name the failure, and quantify the error. Use the taxonomy from Session 7: silent fan-out, a dimension join acting as a filter, wrong grain, date logic, or a filter that quietly changes the answer.
- **One reconciliation.** Show that a total computed from your star matches the same total computed from the source extract, or account for the difference exactly.

6 · Disclosure note — `disclosure.md`

Half a page. What the agent did, what you did, where you overrode it, and one place where it was wrong. A submission without a disclosure note is incomplete and will not be graded.

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5 · The checkpoint, Thursday October 1

Bring to class, and submit on MyCourses that day: page two of your memo — the framing and your five decisions, in draft — your grain statements, and your bus matrix. No code.

WHY THIS EXISTS

It is graded on completion, not quality, and it is there to catch a grain decision you made by accident before you build four weeks of work on top of it. The students who write the grain sentence *after* the query are the students who rewrite the project in the last week.

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6 · Working with an agent

The workflow from class applies, and it is the workflow I expect to see in your notebook: write the steps as comments first, have the agent fill them in one at a time, then check what came back. The comments are the part that is yours.

What it will get wrong here

Two failures are near-certain on this dataset. It will join orders to order lines and then sum an order-level column. And it will assume a column means what its name suggests. Both produce a number that looks right.

Where it is genuinely useful

Boilerplate, build scripts, a first draft of a metric entry, and a second opinion on a query you already wrote. Read the data dictionary before you accept any of it, or you will not notice what it got wrong.

7 · How this is graded

COMPONENT	POINTS
Framing and recommendation — is there a decision, and does the memo serve it	20
Model — grain, bus matrix, star, and whether the build script runs	20
Metric definitions — complete, unambiguous, honest about what was excluded	20
Verification log — traced numbers, audited query, reconciliation	25
Analysis notebook — reproducible, readable, the right work in the right layer	15
Total	**100**

What earns the top of a band: a defensible choice stated plainly, with the alternative named and its consequence quantified. **What loses points fastest:** a number nobody can trace, a definition two people would read two ways, and a grain that was decided by whatever the query happened to return.

I may ask you to reproduce one of your traced numbers for me. Per the syllabus, "the agent produced it" is not an answer to a question about your own work.

8 · Logistics

- **Late work** is penalized 10% for every twelve-hour block, per the syllabus.
- **Questions** go to the MyCourses discussion board rather than email, so that everyone sees the answer. Questions about what a column means, I will answer. Questions about which attribution rule to use, I will not — that is the project.
- **Office hours** are the right place to test a grain decision before you build on it.